

1. Math via the Base14 path

`#set text(font: "Helvetica")` routes through Mosaic's Base14 encoder, which layers a per-document `/Differences` array on top of `WinAnsiEncoding` to reach the AFM glyphs beyond WinAnsi. The ten math symbols Adobe ships in every Core 14 Latin AFM are available without embedding any font — the outlines come from the reader's built-in Helvetica.

Inequalities: $a \leq b$, $x \geq y$, $p \neq q$.

Calculus: $\partial f / \partial x = 0$, $\sum a = 1$, $\sqrt{n} \geq 0$.

Geometry: $\triangle ABC$, optional steps marked $\diamond$.

The minus operator ($a - b$) uses U+2212, distinct from the ASCII hyphen-minus ($a - b$). The fraction slash ($1/2$) uses U+2044.

1.1. Coverage

This path covers exactly ten math glyphs — `lessequal`, `greaterequal`, `notequal`, `partialdiff`, `summation`, `radical`, `minus`, `Delta`, `lozenge`, `fraction` — the AGL- named glyphs every Core 14 Latin AFM ships. Symbols outside that list (infinity, forall, exists, element-of, nabla, product, oplus, otimes, sub/superscripts, calligraphic and blackletter alphabets) require font embedding and are unreachable through the `/Differences` path. The eventual fix is bundling Noto Sans Math with per-glyph font fallback so math codepoints route through it while body text stays in the primary family.

1.2. Caveat

Glyph coverage past WinAnsi depends on each PDF reader's chosen Helvetica substitute, not just on Mosaic emitting a correct `/Differences` declaration. Even within the poppler family, different installs (GNOME Papers vs Evince vs zathura) can pick different substitute fonts via `fontconfig` and end up with different glyph coverage — `lozenge` ($\diamond$) in particular is prone to being absent. Adobe Reader has separately been observed to mis-render `/Differences`-mapped glyphs by splicing in spacing-substitution composition (Romanian comma-below in particular). If you target a specific reader and rely on any specific glyph from the AGL math subset, verify the rendering on that reader before shipping.